

RCIM Model-Bank Reproduction Forward DT Paper-Faithful Search Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

- campaign name: `track1_bidirectional_paper_faithful_grid_search_campaign_2026-05-04_12_26_30__forward_dt_search`
- parent planning report: `doc/reports/campaign_plans/track_1/exact_paper/2026-05-04-12-13-07_track1_paper_faithful_search_protocol_and_campaign_replacement_plan_report.md`
- campaign output directory: `output/training_campaigns/track1/exact_paper/bidirectional_paper_faithful_grid_search/track1_bidirectional_paper_faithful_grid_search_campaign_2026-05-04_12_26_30__forward_dt_search`
- queue size: `1`
- direction scope: `forward`
- family scope: `DT`
- workflow stage: `search`
- launch mode: `remote_operator_launcher`
- remote host alias: `xilab-remote`

Completion Summary

- the subset exact-paper campaign completed the full `1/1` queue successfully;
- remote artifact sync-back completed successfully and materialized the local log, validation bundle, and validation report;
- the run executed the paper-faithful search path with repository-aligned historical `cross_validate(...)` replay at both wrapper and target levels;
- the subset closeout materialized campaign-level `campaign_leaderboard.yaml`, `campaign_best_run.yaml`, and `campaign_best_run.md` bookkeeping artifacts;
- the parent `20 -run paper-faithful RCIM Model-Bank Reproduction` campaign remains canonically `cancelled` and is not reopened by this subset closeout.

Campaign Best Run

- run instance id: 2026-05-08-17-08-23__track1_paper_faithful_grid_search_forward_dt_campaign_validation
- run name: track1_paper_faithful_grid_search_forward_dt
- direction label: forward
- winning family: DT
- winning estimator: DecisionTreeRegressor
- winning mean component MAE: 0.095493
- winning mean component RMSE: 0.186076
- winning mean component MAPE: 40.5591%
- best parameter source: grid_search
- best parameters: {'estimator__criterion': 'squared_error', 'estimator__max_depth': 14, 'estimator__max_leaf_nodes': 23, 'estimator__min_samples_split': 5}

Selection rule:

- Primary metric: winning_mean_component_mae_asc
- First tie-breaker: winning_mean_component_rmse_asc
- Second tie-breaker: winning_mean_component_mape_percent_asc
- Third tie-breaker: run_name

Search Protocol Summary

Item	Value
Grid search candidates	300
Grid search CV folds	5
Estimated grid-search CV fits	1500
Historical wrapper cross_validate(...) folds	10
Historical target-level cross_validate(...) folds	10 per target
Target count	19
Random seed	0

Validation Outcome

Metric	Value
Mean component MAPE %	40.559
Mean component MAE	0.095493
Mean component RMSE	0.186076
Train rows	678
Validation rows	194
Test rows	97
ONNX exports	19
ONNX export failures	0

Historical Replay Outcome

Scope	Mean MAE	Mean RMSE	Mean MAPE %
Wrapper <code>cross_validate(...)</code> replay	0.082761	0.196496	47.602
Held-out test evaluation	0.095493	0.186076	40.559

Linked Artifacts

- campaign log:
`output/training_campaigns/track1/exact_paper/bidirectional_paper_faithful_grid_search/track1_bidirectional_paper_faithful_grid_search_campaign_2026-05-04_12_26_30__forward_dt_search/logs/001_track1_paper_faithful_grid_search_forward_dt.log`
- remote wrapper log: `.temp/remote_training_campaigns/2026-05-08-17-07-22_track1_bidirectional_paper_faithful_grid_search_campaign_2026_05_04_12_26_30__forward_dt_search/remote_training_campaign.log`
- validation report: `doc/reports/analysis/validation_checks/2026-05-08-17-10-02_paper_rei_57e160d8_track1_paper_faithful_grid_s_bd96d56f_original_dataset_exact_model_bank_report.md`
- validation summary:
`output/validation_checks/paper_reimplementation_rcim_original_dataset_exact_model_bank/2026-05-08-17-08-23__track1_paper_faithful_grid_search_forward_dt_campaign_validation/validation_summary.yaml`
- best-parameter summary:
`output/validation_checks/paper_reimplementation_rcim_original_dataset_exact_model_bank/2026-05-08-17-08-23__track1_paper_faithful_grid_search_forward_dt_campaign_validation/best_parameter_summary.yaml`
- model bundle:
`output/validation_checks/paper_reimplementation_rcim_original_dataset_exact_model_bank/2026-05-08-17-08-23__track1_paper_faithful_grid_search_forward_dt_campaign_validation/paper_family_model_bank.pkl`
- campaign leaderboard:
`output/training_campaigns/track1/exact_paper/bidirectional_paper_faithful_grid_search/track1_bidirectional_paper_faithful_grid_search_campaign_2026-05-04_12_26_30__forward_dt_search/campaign_leaderboard.yaml`
- campaign best run YAML:
`output/training_campaigns/track1/exact_paper/bidirectional_paper_faithful_grid_search/track1_bidirectional_paper_faithful_grid_search_campaign_2026-05-04_12_26_30__forward_dt_search/campaign_best_run.yaml`
- campaign best run Markdown:
`output/training_campaigns/track1/exact_paper/bidirectional_paper_faithful_grid_search/track1_bidirectional_paper_faithful_grid_search_campaign_2026-05-04_12_26_30__forward_dt_search/campaign_best_run.md`
- parent active campaign state: `doc/running/active_training_campaign.yaml`

Closeout Notes

- this is a subset closeout for the successful `forward + DT + search` bundle launched through the family-stage exact-paper wrapper;
- it is not a canonical full closeout of the parent paper-faithful `20`-run RCIM Model-Bank Reproduction campaign;
- no RCIM Model-Bank Reproduction benchmark refresh, master-summary refresh, or paper-reference archive refresh is performed here because the parent campaign remains interrupted and incomplete;
- the subset closeout is intended to preserve auditable evidence for the new launcher, logging, and paper-faithful search protocol on one completed family-direction run.